

Formative activity

1 Context of the activity

An earthquake is the sudden release of energy accumulated within the Earth's tectonic plates. This generates vibrations that propagate in the form of seismic waves. Seismologists use scales such as the Richter scale to measure the strength of these phenomena. Table 1 shows the dates, origin and magnitude of some earthquakes that have happened in Mexico.

Date	Place	Magnitude
28 mar 1787	Oaxaca coastline	8.6
19 sep 1985	Michoacán / CDMX	8.1
7 sep 2017	Gulf of Tehuantepec	8.2
19 sep 2017	Morelos-Puebla	7.1
6 jul 1964	Tierra Caliente	7.2-7.3
23 ago 1965	Oaxaca	7.3-7.5
20 mar 2012	South of Ometepepec, Guerrero	7.5-7.7

Table 1: Major earthquakes in Mexico and the energy they release

The Richter scale is based on a logarithmic function that relates the amplitude of the waves to a numerical magnitude. For instance, an earthquake measuring 4.0 on the Richter scale may be perceived as a slight vibration indoors, whereas a magnitude of 6.0 could result in moderate damage to vulnerable structures. In contrast, an earthquake measuring 8.0 or higher, such as the one that struck Michoacán in 1985, can be devastating.

Using mathematical models such as the Gutenberg-Richter equation enables us to calculate the energy released during an earthquake and make quantitative comparisons between events. According to Gutenberg-Richter's law, which describes the seismic distribution of magnitude and frequency in Mexico, the typical constants adjusted for the central region of the country¹ are as follows:

$$\log [N(M)] = 4.54 - 0.55M. \quad (1)$$

Where $N(M)$ represents the number of earthquakes at the same magnitude, M , as defined on the Richter scale.

These models are essential for understanding the relationship between earthquake magnitude and the amount of energy released. For example, an increase of one unit on the Richter scale implies that approximately 32 times more energy is released. Therefore, a 7.0 magnitude earthquake

¹Equation retrieve from the journal of the Mexican geophysical union. Link to download the document

releases energy equivalent to the explosion of 500 000 tons of TNT, whereas a 5.0 magnitude earthquake releases energy comparable to 500 tons of TNT. The algebraic expression that relates the energy released by an earthquake with the following equation,

$$\log [E] = 4.8 + 1.5M. \quad (2)$$

Where M is the magnitude of the earthquake in the Richter scale and E is the energy in joules (force times distance). In this way, mathematics helps us not only to quantify natural phenomena, but also to prepare for their possible effects.

2 Activity

In this activity, we will use the given information to develop an understanding of how functions can be used to quantify natural phenomena. To do this, we will carry out the following:

- Find the algebraic expression for the difference of energy between different magnitudes.
- Calculate the energy of an earthquake in terms of the frequency of earthquakes of the same magnitude in Mexico.
- Graph the functions and analyze the order of magnitude of the energy.

2.1 Algebraic tools

Laws of exponents

$$\begin{aligned} a^r \cdot a^s &= a^{r+s}, & (a^r)^s &= a^{rs} \\ \frac{a^r}{a^s} &= a^{r-s}, & (ab)^r &= a^r b^r \\ \left(\frac{a}{b}\right)^r &= \frac{a^r}{b^r}, & b &\neq 0 \end{aligned}$$

Laws of Logarithms

$$\begin{aligned} \log_a (AB) &= \log_a (A) + \log_a (B) \\ \log_a \left(\frac{A}{B}\right) &= \log_a (A) - \log_a (B) \\ \log_a (A^C) &= C \log_a (A) \end{aligned}$$

Inverse functions property

$$(f \circ f^{-1})(x) = x = (f^{-1} \circ f)(x) \quad \rightarrow \quad \log_a (a^x) = x = a^{\log_a (x)}$$